

# BLUESTOCK FINTECH

## Mutual Fund Analytics

Capstone Project I

## DEMO VIDEO SCRIPT

Full narration guide for the 4-5 minute project walkthrough

|                |                                                    |
|----------------|----------------------------------------------------|
| Team Members:  | <b>asthananeha2015 / bsoham2704 / Bubai Das</b>    |
| Project:       | <b>Capstone Project I -- Mutual Fund Analytics</b> |
| Target Length: | <b>4 to 5 minutes (screen recording)</b>           |
| Tools Covered: | <b>Python   Jupyter   SQLite   PowerPoint</b>      |
| Date:          | <b>July 2026</b>                                   |

*Recommended tool: OBS Studio / Loom / Windows Game Bar (Win + G)*

*Resolution: 1920 x 1080 | Format: MP4*

# VIDEO OVERVIEW & PREPARATION CHECKLIST

## Timeline at a Glance

| Time        | Section                | What to Cover                                   |
|-------------|------------------------|-------------------------------------------------|
| 0:00 - 0:30 | Introduction           | Greet, state project name, team, and tech stack |
| 0:30 - 1:45 | Dashboard Walkthrough  | Open PPTX, walk 4 pages, explain KPIs & charts  |
| 1:45 - 3:00 | Jupyter Notebooks      | EDA, Performance Analytics, Advanced Analytics  |
| 3:00 - 3:40 | Fund Recommender (CLI) | Run recommender.py --risk Moderate in terminal  |
| 3:40 - 4:20 | SQLite Database        | DB Browser, run queries, show star schema       |
| 4:20 - 4:45 | Closing & Summary      | Recap key findings, thank panel                 |

## Before You Press Record -- Open These Files

- PPT\_Slides/bluestock\_mf\_dashboard.pptx
  - Source\_Code/notebooks/EDA\_Analysis.ipynb
  - Source\_Code/notebooks/Performance\_Analytics.ipynb
  - Source\_Code/notebooks/Advanced\_Analytics.ipynb
  - Datasets/bluestock\_mf.db
  - Source\_Code/sql/queries.sql
  - Terminal / Command Prompt
- <- Open in Microsoft PowerPoint, go to Slide Show view
  - <- Open in Jupyter Notebook (run all cells first)
  - <- Open in Jupyter Notebook (run all cells first)
  - <- Open in Jupyter Notebook (run all cells first)
  - <- Open in DB Browser for SQLite
  - <- Open in a text editor beside DB Browser
  - <- cd to Source\_Code/scripts/ before recording

TIP: Run all notebook cells BEFORE recording so output is already visible. This saves time and avoids errors on camera.

## SECTION 1 INTRODUCTION

Duration: 0:00 - 0:30 | Screen: Desktop or title slide

0:00

Hello everyone! My name is Neha Asthana, and I am presenting the Bluestock Fintech Mutual Fund Analytics project -- Capstone Project I. This project was built by our team: myself, Soham Biswas, and Bubai Das.

0:08

In this project, we performed an end-to-end analytics pipeline on 40 Indian mutual fund schemes across 10 fund houses -- covering data ingestion, cleaning, SQL schema design, exploratory analysis, performance metrics, advanced risk analytics, and finally a 4-page interactive dashboard.

0:20

Our tech stack is entirely Python-based -- using pandas, NumPy, Matplotlib, Seaborn, Plotly, SQLAlchemy with SQLite, and python-pptx for the dashboard. All code is version-controlled on GitHub.

>> ACTION: Show the GitHub repository briefly OR show the submission folder structure on screen

The submission is organised into five folders: Source Code, Datasets, Documentation, PPT Slides, and Demo Video. Let me now walk you through the key deliverables.

SECTION 2DASHBOARD WALKTHROUGH

Duration: 0:30 - 1:45 | File: PPT\_Slides/bluestock\_mf\_dashboard.pptx

0:30

>> ACTION: Switch to PowerPoint -- bluestock\_mf\_dashboard.pptx. Start Slide Show (F5). Go to Slide 1 (Title Slide).

Let us start with the dashboard. I have opened our PowerPoint presentation -- bluestock\_mf\_dashboard.pptx -- which contains five slides built using python-pptx with our Bluestock brand theme.

Slide 2 -- Industry Overview (0:38)

>> ACTION: Advance to Slide 2 (Industry Overview). Highlight the 4 KPI cards at the top.

| TOTAL INDUSTRY AUM | ANNUAL SIP INFLOWS | TOTAL FOLIOS | TOTAL SCHEMES |
|--------------------|--------------------|--------------|---------------|
| Rs.81L Cr          | Rs.31K Cr          | 26.12 Cr     | 1,908         |

Slide 2 is our Industry Overview. At the top, you can see four KPI cards. Total Industry AUM stands at Rupees 81 Lakh Crore as of March 2026 -- reflecting the massive scale of Indian mutual fund industry. Annual SIP inflows were Rupees 31,000 Crore for FY2025, showing strong retail investor participation. There are 26.12 Crore total folios and 1,908 active schemes across all AMCs.

The line chart below shows the AUM growth trend from 2022 to 2026 for our sample of 10 fund houses -- you can clearly see a consistent upward trend. The horizontal bar chart on the right ranks each AMC by their AUM -- with SBI Mutual Fund leading, followed by HDFC and ICICI Prudential.

Slide 3 -- Fund Performance (1:00)

>> ACTION: Advance to Slide 3 (Fund Performance). Point to the scatter plot.

Slide 3 shows Fund Performance analytics. The scatter plot on the left maps 3-year annualised return on the X-axis against standard deviation -- that is, risk -- on the Y-axis. Each bubble's size represents the fund's AUM. The ideal position is the top-left: high return with low risk. You can see Large Cap and Liquid funds clustered in the lower-left, while Small Cap funds appear high-right with greater risk and greater return potential.

>> ACTION: Point to the scorecard table on the right side.

The fund scorecard table ranks the top 10 funds by Sharpe ratio -- this measures risk-adjusted return. The higher the Sharpe, the better the return per unit of risk taken. We also show 3-year return, expense ratio, and the risk grade assigned by SEBI. The NAV chart at the bottom right shows NAV growth for the top 3 equity funds indexed against the NIFTY 100 benchmark.

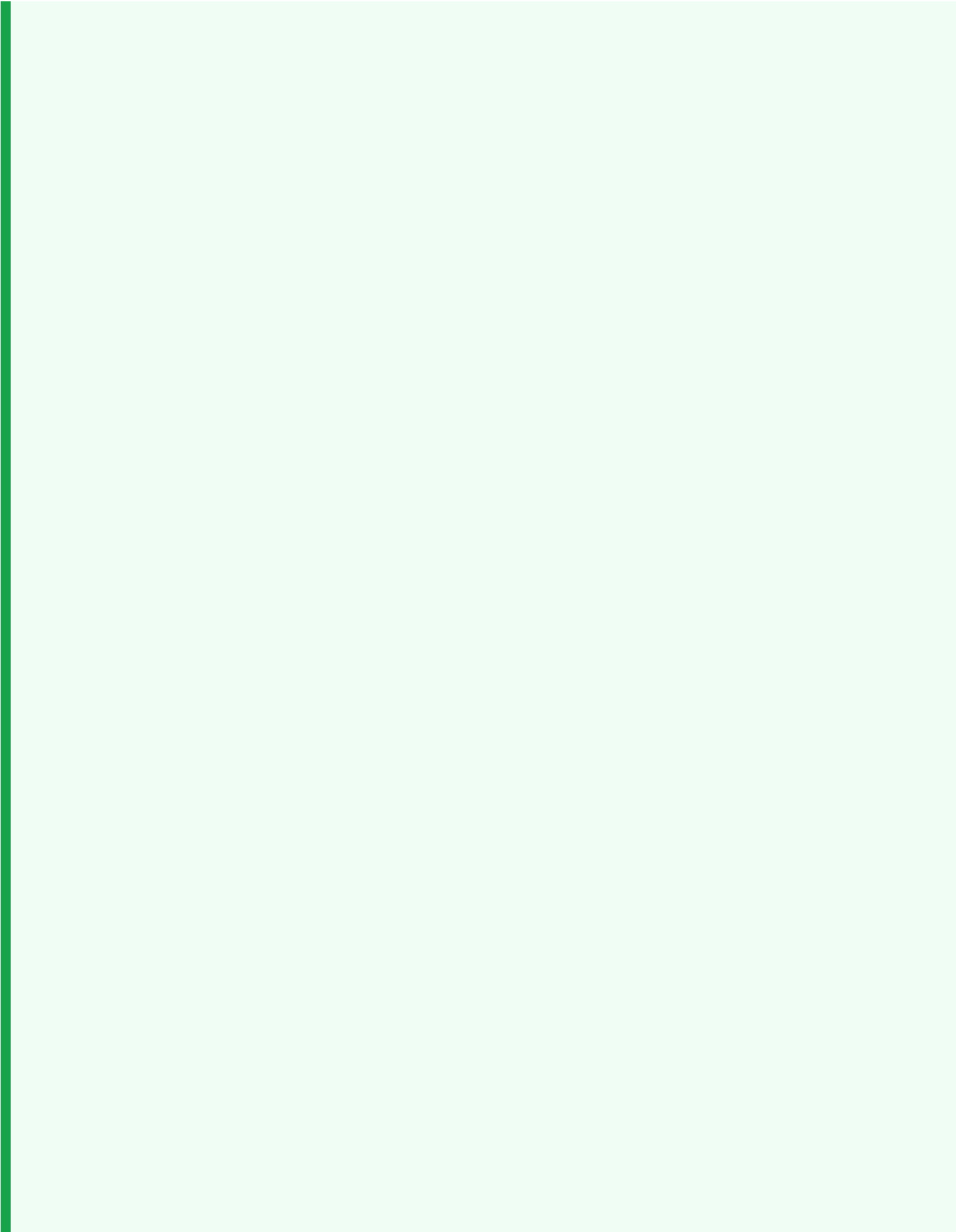
Slide 4 -- Investor Analytics (1:18)

>> ACTION: Advance to Slide 4 (Investor Analytics).

Slide 4 covers Investor Analytics. The top-left bar chart shows transaction volume by state -- Maharashtra and Karnataka lead, reflecting higher financial literacy in metro cities. The donut chart breaks down total investment by type: SIP, Lumpsum, and Redemption. SIP dominates, confirming the systematic investment culture among Indian retail investors.

The bottom-left bar shows average SIP ticket size by age group -- interestingly, the 56-plus age group has the

highest average SIP amount, suggesting retirement-focused wealth building. The bottom-right line chart tracks monthly transaction volume, showing growth from January 2024 onwards.



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### Slide 5 -- SIP & Market Trends (1:33)

>> ACTION: Advance to Slide 5 (SIP & Market Trends).

Slide 5 is SIP and Market Trends. The large dual-axis chart at the top plots monthly SIP inflows as blue bars and the NIFTY 50 index as an orange line from 2022 to 2026. A key insight here: even during the 2024 market correction when NIFTY 50 dipped, SIP inflows continued to grow -- showing that systematic investors stay disciplined during volatility.

The heatmap shows net category inflows for the last 12 months -- darker orange means higher inflows. Mid Cap and Flexi Cap funds dominate inflows. The bar chart on the right shows the top 5 fund categories by net inflows for FY2025 -- Mid Cap leads with Rupees 22,000-plus Crore.

## SECTION 3 JUPYTER NOTEBOOKS

Duration: 1:45 - 3:00 | Source\_Code/notebooks/

1:45

>> ACTION: Switch to browser with Jupyter Notebook. Open EDA\_Analysis.ipynb.

Now let us look at the Jupyter notebooks which form the analytical core of this project. I have three key notebooks to show you.

### EDA\_Analysis.ipynb (1:48)

>> ACTION: Scroll to Cell 1 -- the imports and data loading. Then scroll to the first chart output.

EDA Analysis notebook covers 15 charts generated during Day 3. Let me highlight three key ones. First -- the NAV trend chart for Large Cap funds shows that all funds trended upward from 2022 to 2026, with a visible correction in early 2024 followed by recovery.

>> ACTION: Scroll to the AUM by AMC bar chart output.

Second -- the AUM growth chart by fund house confirms SBI Mutual Fund as the largest with over 6 Lakh Crore under management. Third -- the correlation matrix shows NAV returns across funds, revealing that Large Cap funds are highly correlated with each other but less so with Small Cap funds -- a useful diversification insight.

*TIP: You do not need to explain every chart. Pick the 2 most interesting visuals from each notebook to keep the demo focused.*

### Performance\_Analytics.ipynb (2:15)

>> ACTION: Switch to Performance\_Analytics.ipynb. Scroll to the fund scorecard table output.

The Performance Analytics notebook computes all standard fund metrics: CAGR for 1, 3, and 5 years; Sharpe and Sortino ratios; Alpha and Beta against the NIFTY 100 benchmark; and maximum drawdown.

>> ACTION: Scroll to the fund scorecard CSV output cell -- show the top 5 rows.

The fund scorecard ranks all 40 funds using a weighted composite score: 30 percent for 3-year return rank, 25 percent for Sharpe rank, 20 percent for Alpha rank, 15 percent for expense ratio, and 10 percent for max drawdown. Our top scorer is ICICI Prudential Midcap Fund Regular with a score of 84.5 out of 100. We also computed Alpha and Beta using OLS regression -- funds with Alpha above 1 percent are generating genuine excess returns beyond what the market explains.

### Advanced\_Analytics.ipynb (2:38)

>> ACTION: Switch to Advanced\_Analytics.ipynb. Go to the VaR/CVaR section.

The Advanced Analytics notebook covers Day 5 risk metrics. Section 1 shows Historical VaR and CVaR at 95 percent confidence. SBI Small Cap Direct has the worst daily VaR of negative 2.69 percent -- meaning on the worst 5 percent of trading days, investors could lose over 2.69 percent of NAV. ICICI Prudential Liquid Fund shows near-zero VaR, confirming its capital-preservation role.

>> ACTION: Scroll to the Rolling Sharpe chart output.

The rolling 90-day Sharpe chart for the top 5 equity funds shows how risk-adjusted performance evolved over time. The 2024 correction is clearly visible -- Sharpe ratios dropped to near zero or negative for all funds. Mirae Asset Large Cap recovered fastest post-correction.

>> ACTION: Scroll to SIP Continuity results.

Our SIP continuity analysis found that 97.8 percent of investors with 6 or more SIP transactions show average gaps greater than 35 days -- suggesting a quarterly or bi-monthly SIP pattern in this dataset. Only 2.2 percent follow strict monthly SIPs.



## SECTION 4 FUND RECOMMENDER SYSTEM

Duration: 3:00 - 3:40 | File: Source\_Code/scripts/recommender.py

3:00

>> ACTION: Switch to terminal / command prompt. Navigate to Source\_Code/scripts/

Let me now demonstrate the fund recommender system. This is a standalone Python script that recommends the top 3 funds based on an investor's risk appetite -- Low, Moderate, or High -- using the Sharpe ratio as the primary ranking metric filtered by SEBI risk grade.

>> ACTION: Type and run the following command in terminal:

```
python recommender.py --risk Low
```

For a Low risk appetite, the recommender returns Liquid and Gilt funds -- these have near-zero VaR and stable NAV, perfect for capital preservation.

>> ACTION: Run again with Moderate risk:

```
python recommender.py --risk Moderate
```

For Moderate risk, we get Large Cap and Flexi Cap funds. Notice the higher Sharpe ratios -- these funds deliver better risk-adjusted returns while staying within a manageable volatility range.

>> ACTION: Run once more with High risk:

```
python recommender.py --risk High
```

For High risk appetite -- typically younger investors with a longer time horizon -- the recommender suggests Mid Cap and ELSS funds. These show the highest 3-year CAGR in our universe, though with greater drawdown risk.

*TIP: If the terminal is hard to read on screen, increase font size to 14pt before recording.*

## SECTION 5 SQLITE DATABASE

Duration: 3:40 - 4:20 | File: Datasets/bluestock\_mf.db

3:40

>> ACTION: Switch to DB Browser for SQLite with bluestock\_mf.db already open.

Now let us look at the data layer. We designed a star schema in SQLite with 6 tables: dim\_fund, dim\_date, fact\_nav, fact\_transactions, fact\_performance, and fact\_aum. This enables efficient time-based joins and slice-and-dice queries across fund, date, and investor dimensions.

>> ACTION: Click 'Browse Data' tab. Show the dim\_fund table -- scroll through a few rows.

The dim\_fund dimension table stores all 40 fund master records including AMFI code, fund house, scheme name, category, SEBI category code, expense ratio, benchmark, and risk grade.

>> ACTION: Click 'Execute SQL' tab. Paste and run this query from queries.sql:

```
SELECT f.scheme_name, p.sharpe_ratio, p.return_3yr_pct
```

```
FROM fact_performance p JOIN dim_fund f ON p.amfi_code = f.amfi_code
```

```
WHERE f.category = 'Equity' ORDER BY p.sharpe_ratio DESC LIMIT 5;
```

This query joins the performance fact table with the fund dimension to show the top 5 equity funds ranked by Sharpe ratio -- a key analytical query that would power any fund comparison feature in a production system.

>> ACTION: Show one more quick query -- AUM by fund house from fact\_aum:

```
SELECT f.fund_house, SUM(a.aum_crore) AS total_aum
```

```
FROM fact_aum a JOIN dim_fund f ON a.amfi_code = f.amfi_code
```

```
GROUP BY f.fund_house ORDER BY total_aum DESC;
```

This aggregation query shows total AUM per fund house -- the same data that powers the AUM by AMC bar chart in our dashboard.

## SECTION 6 CLOSING & KEY FINDINGS

Duration: 4:20 - 4:45 | Screen: Dashboard or Submission Folder

4:20

>> ACTION: Switch back to the dashboard PPTX or show the submission folder tree.

To wrap up, let me summarise the five key findings from this project.

### 1 Indian MF Industry Growth

Industry AUM grew consistently from 2022 to 2026 despite the 2024 correction. SIP inflows remained resilient even when NIFTY 50 dipped -- demonstrating the maturity of Indian retail investors.

### 2 Risk-Return Profile

Small Cap funds offer the highest 3-year CAGR (above 28 percent) but carry the worst daily VaR of minus 2.69 percent. Liquid funds deliver near-zero risk with stable 6-7 percent annualised returns -- suitable for emergency funds.

### 3 Fund Scorecard Winner

ICICI Prudential Midcap Fund Regular scored 84.5 out of 100 on our composite scorecard, balancing strong 3-year returns, a Sharpe ratio above 1.2, and a competitive expense ratio of 1.4 percent.

### 4 Investor Behaviour Insight

2025 cohort investors bring Rupees 2,500 higher average SIP tickets than 2024 cohort investors -- suggesting newer investors are entering with greater financial confidence and awareness.

### 5 Portfolio Concentration Risk

Axis Bluechip has the highest sector HHI of 2,968 -- heavily concentrated in a single sector. UTI Mid Cap is the most diversified with HHI of 1,241 across 9 sectors. Investors should prefer low-HHI funds for true diversification.

4:38

This project demonstrates a complete data science pipeline -- from raw CSV ingestion to a production-ready analytics dashboard -- applied to a real-world financial domain. All code is open-source and available on our GitHub repository. Thank you for watching!

>> ACTION: Stop recording. Save the video as MP4 and upload to submission/Demo\_Video/ on Google Drive.

### POST-RECORDING CHECKLIST

- ☐ Video is 4-5 minutes long and saved as MP4 (1920x1080)
- ☐ Upload video to submission/Demo\_Video/ in Google Drive
- ☐ Make Google Drive folder public: Anyone with the link -> Viewer
- ☐ Paste the folder link in the submission form and click Submit